



A shaping two-stage anomaly data recovery method based on multi-norm joint optimization under energy internet

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ABSTRACT

With the deep integration of energy networks and information communication technology, the involvement of hardware and environmental factors intensifies the challenges to data security. Prior research focuses predominantly on interpolating random missing data under the assumption of no data damage, without providing strategies for addressing continuous data loss. To overcome, we propose a novel two-stage data recovery method, that includes three-fold ideas: (1) it introduces the matrix shaping differentiation strategy to realize the differentiated recovery of random and continuous missing data; (2) it designs a two-stage data recovery approach, in which the first stage constructs a random missing data recovery objective function by eliminating the consecutive missing data columns, and the second stage performs temporal pre-interpolation backfilling on the eliminated columns and introduces the temporal correlation to construct the temporal smoothing objective function; and (3) it develops an alternating direction method of multipliers-based iterative solution algorithm. Experimental results on four public datasets demonstrate that our model is superior to five state-of-the-art and classical methods in terms of random missing error rate, random column missing error rate, root mean square error, and mean absolute error.

1. Introduction

To address the continuously growing energy demand and environmental sustainability challenges, a new energy supply model has emerged: the Energy Internet [1]. The Energy Internet aims to utilize advanced information and communication technologies to interconnect various stages of energy production, transmission, distribution, and usage, enabling a more efficient and sustainable smart energy management. Unlike traditional centralized energy systems, the Energy Internet encourages the integration of distributed energy resources, smart grids, energy storage, and energy management technologies to achieve energy peer-to-peer exchange and shared networks [2]. A schematic view of Energy Internet is illustrated in the left panel of Fig. 1.

In the processes of power generation, transmission, storage, and intelligent management within the Energy Internet, a substantial volume of intelligent energy data [3,4] has emerged. This encompassing dataset includes system monitoring data, operational records, power transmission specifics, weather patterns, commercial data, and more [5–7]. The diverse and abundant data, sourced from various points across generation, grid, load, and storage stages, is acquired through sensing mechanisms. Leveraging precise sensor data such as load currents and

temperature rise information supports critical energy equipment like transformers and casing, aiding in status assessment and fault analysis. Consequently, this approach diminishes major incident risks like fires, ensuring the secure and dependable operation of these vital devices. Sensors play a crucial role in monitoring factors such as wind speed and solar radiation for renewable energy sources like solar and wind power. This monitoring optimizes energy output and ensures maximal utilization of renewable resources. Power prediction in new energy generation, based on comprehensive measurement data and advanced algorithms, enables precise decision-making regarding load trends, behaviors, demands, and outcomes, effectively addressing the stochastic fluctuations and uncertainties on both the supply and demand sides [8]. Additionally, continuous monitoring of critical transmission channels for factors such as wildfires, typhoons, and icing conditions through external environmental data enables round-the-clock, comprehensive status checks and risk alerts. Therefore, sensor-measured data is a fundamental cornerstone within the information support system of the Energy Internet [9].

As information and communication become more integrated into the Energy Internet, cybersecurity threats are on the rise. Measurement

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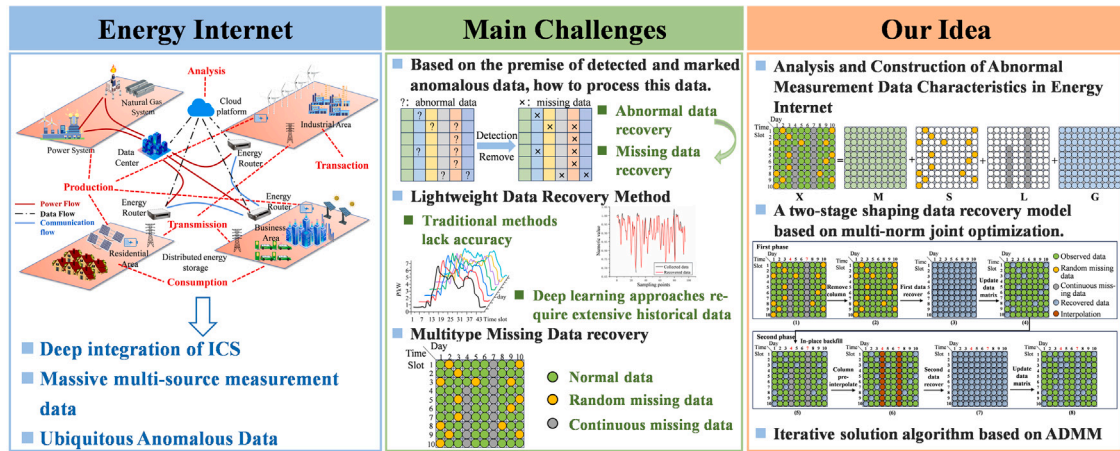


Fig. 1. Energy internet background, the three challenges, and our main ideas to MOSDR.

data is vulnerable to network attacks during collection and transmission. Attackers may tamper with measurement data, leading to energy supply disruptions or unfair distribution, compromising the reliability and fairness of the energy system. This could cause power shortages, which pose severe risks to grid stability and the economy [10]. Moreover, in harsh and unattended environments, sensor nodes may malfunction, resulting in noise, errors, data loss, or redundancy. Therefore, ensuring accurate and comprehensive measurement data is critical for the proper functioning of the Energy Internet system. How to accurately and quickly recover data after anomalous data occurs is urgently needed.

The recovery of abnormal data typically involves two stages: abnormal data detection and data restoration. Currently, numerous methods exist for detecting abnormal data, such as machine learning-based approaches and deep neural network-based methods [11]. Once abnormal data has been accurately detected and labeled, whether numerical anomalies or data gaps, a unified preprocessing strategy is employed in this study. All such abnormal data is uniformly labeled as missing data for subsequent processing. The common methods for missing data recovery include various interpolation techniques such as K-nearest neighbors (KNN) [12] and Delaunay Triangulation (DT) [13]. However, interpolation methods rely solely on local data information, leading to inaccuracies in recovered data, especially in cases of continuous or structural data absence. With the advancement of artificial intelligence technology, learning-based methods have been applied to the field of missing data recovery. For example, recovery frameworks like Hierarchical Long Short-Term Memory (ST-HLSTM) [14] and data recovery methods based on General Regression Neural Networks (GRNN) [15] have emerged. Although deep neural network models have shown exceptional performance in extensive datasets, data-driven modeling approaches require substantial historical training data. However, wireless sensor networks often collect only limited time-series data due to the high cost of sensor probes, poor operating conditions, or limited power supply. Therefore, traditional deep learning techniques may not be effective for these systems. In recent years, methods based on matrix completion have demonstrated promising performance in the field of missing data recovery within the Energy Internet. Leveraging the low-rank characteristics of measurement data, employing algorithms such as Singular Value Thresholding (SVT) [16] or matrix factorization [17] to minimize matrix rank has proven effective in recovering a significant portion of missing information. Given the presence of noise in real-world environments, Robust Principal Component Analysis (RPCA) has exhibited its advantage in handling noise-contaminated data recovery. It decomposes data into low-rank and sparse matrices, effectively separating abnormal data from background data [18]. However, these conventional matrix completion methods often assume missing data to be random and uniform, whereas in reality, missing data tend to exhibit

structural patterns, such as sensor failures or prolonged continuous gaps due to DDoS attacks. Additionally, although existing methods [19] can leverage spatiotemporal correlations and rank constraints for data recovery, obtaining topological information about sensor nodes in dynamic and complex Energy Internet environments is often challenging, demanding high practical usability conditions.

Urged by the situation, we propose a novel two-stage data recovery method based on multi-norm joint optimization, termed Multinorm Optimized Structured Data Recovery (MOSDR). Its core ideas, shown in the right panel of Fig. 1, include three-folds: (1) data characterization and matrix construction for multi-scenario measurement data of EI; (2) the first stage of matrix recovery is carried out by firstly removing long-time consecutively missing data columns and constructing a random missing data matrix recovery model by utilizing the shaping data; and (3) the missing columns deleted in the first stage are interpolated in the mean, and using the temporal correlation of the measurement data, the second stage missing data matrix recovery model is constructed to further improve the data recovery accuracy for the data recovered in the first stage.

More specifically, Our major contributions are summarized as follows:

- By analyzing the characteristics of EI measurement anomaly data, we realize a multi-matrix superposition decomposition of the observation data matrix. This decomposition provides the theoretical foundation for our proposed method.
- We introduce the MOSDR method, which enhances recovery accuracy by accounting for matrix shaping and temporal correlation, and by performing data recovery in phases to address the simultaneous existence of different types of missing data.
- We design an iterative solution algorithm based on the Alternating Direction Method of Multipliers (ADMM), which decouples the variables involved in MOSDR and reduces the computational complexity of solving the MOSDR model.
- Experimental results on four open datasets demonstrate that our MOSDR model achieves higher recovery accuracy compared to seven state-of-the-art and classical methods.

The rest of this paper is organized as follows. Section 2 presents the related work. Section 3 gives the preliminary knowledge. Our proposed MOSDR model is elaborated in the Section 4, and Section 5 analyzes the experimental results. Finally, Section 6 concludes the paper.

2. Related work

To date, existing solutions to the data recovery problem fall into three main categories: interpolation-based methods, artificial

intelligence-based methods, and matrix complementation-based methods. This section aims to provide an in-depth discussion of the pros and cons of these approaches, and to illustrate the connection between them and the data recovery model proposed in this paper.

2.1. Interpolation-based methods

Interpolation is the oldest method of recovering missing data. Linear interpolation uses a linear relationship between two points to estimate the value of the intermediate point. It is one of the best interpolation methods when the missing gap is one or two [20]. However, as the missing gap increases, its accuracy decreases significantly, and linear interpolation mainly considers neighboring known data points and may ignore other important information far away from the missing point. Due to the limitations of linear interpolation methods, scholars have explored more advanced techniques, including regression interpolation. Unlike linear interpolation, regression interpolation predicts missing values based on known data points and their relationships with other variables. This method integrates the correlations between multiple variables, providing a comprehensive view of data recovery [21]. The Autoregressive Moving Average (ARMA) model, proposed by P.M.T. Broersen et al. [22], combines the linear processes of Autoregression (AR) and Moving average (MA) to represent a stable time series. It relies on past values and random errors to make predictions. Despite using simplified statistical algorithms, the ARMA model can achieve better results than the log-likelihood function for nonlinear optimization in certain cases. However, it is unable to handle nonstationary trends and seasonal patterns. The Autoregressive Integrated Moving Average (ARIMA) model [23] is used to recover non-stationary time series data in an integrated form of ARMA. However, it may not fully utilize the data collected after missing data, which can result in a less stable model. Additionally, a commonly adopted strategy for replacing missing values is to use the proximity of the measured values. One such method is k-Nearest Neighbors (KNN) [12], which identifies the k closest observations based on a distance matrix and computes a k-weighted average (based on distance weighting) for each observation. Incomplete data is then replaced by the average of corresponding attributes of the k neighbors that contain complete data. The classical local interpolation method is commonly used for low-fidelity estimation situations. However, it may not be suitable for large datasets and can encounter difficulties when dealing with many variables. The Delaunay Triangulation (DT) method [13] treats the collected data as vertices and interpolates the data by reconstructing virtual triangles using them and their global errors.

Remark. From the above, it is clear that traditional interpolation methods primarily address local data characteristics. While effective for handling short-term missing data, their performance is significantly limited for long-term continuous missing data. This is particularly true for energy internet data, which has a high correlation and requires data recovery methods that can capture the deep structure and underlying patterns of the data. Thus, it is not possible for a single interpolation method to accurately recover energy internet measurement data.

2.2. Artificial intelligence-based methods

The advancement of artificial intelligence has expanded the potential for data recovery. Stekhoven et al. [24] proposed a Random Forest based missing data interpolation algorithm that achieves multiple interpolation by averaging multiple unpruned classification or regression trees. Lai et al. [25] used autoencoder for missing value interpolation, the core of this method is to learn how to restore the original input values by minimizing the reconstruction error, by which the complexity of the data can be reduced. Gondara L et al. [26] proposed a multiple estimation model based on an ultra-complete deep denoising autoencoder, which is able to deal with different types, missing modes and

missing proportions of the data. Yoon J et al. [27] used a generative adversarial network (GAN) to estimate missing values. They adjusted the impact of the mean squared error (MSE) loss term and the ratio of the discriminator cue vectors through hyperparameter settings. Tekler Z D et al. [28] proposed an optimal deep learning architecture with a two-step feature selection algorithm for predicting indoor carbon dioxide concentration in different spaces in a building. Karevan Z et al. [29] utilized the long-term dependence capability of LSTM to implement data-driven prediction models in weather forecasting applications and proposed the Transductive LSTM (T-LSTM) method, which exploits local information in time series forecasting by considering samples in the vicinity of the test point, thus achieving better performance in forecasting tasks.

Remark. Although each of the AI-based data recovery methods mentioned above has unique applications and outstanding performance, most require a large amount of training data to ensure model accuracy. Additionally, some methods have high deployment and computational costs in practical applications due to the demand for high-performance computing. Therefore, further research is necessary to investigate lightweight methods for recovering energy Internet measurement data.

2.3. Matrix completion-based methods

Matrix complementation techniques are a popular research topic in data recovery due to the success of compression-aware techniques [30] in signal processing. This process involves filling in the missing entries of a partially observed matrix. Because missing values can be assigned arbitrarily, the problem of matrix complementation is underdetermined when there are no restrictions on the degrees of freedom of the complemented matrices. Therefore, matrix complementation typically aims to find low-rank matrices that match the known values.

The standard matrix-completion problem is widely recognized as NP-hard. However, Candès et al. [31] have shown that matrices consistent with the observed entries can be reconstructed by adjusting the trace-norm. It has been demonstrated that, under strong incoherence conditions, a matrix $M \in \mathbb{R}^{m \times n}$ of rank r can be fully recovered with very high probability as long as the number of observed entries $m \geq Cn^{5/4}r \log n$ (C is a constant). Later, Candès et al. [18] proposed the robust principal component analysis method RPCA, which decomposes a given data matrix M into the sum of a low-rank matrix L and a sparse matrix (outlier matrix) S . This method improves robustness by separating the data into these two matrices. Besides the use of trace-norm instead of matrix rank, matrix decomposition methods by decomposing the matrix to be solved into a matrix product are another common approach. Examples include low-rank matrix fitting (LMaFit) [32], multi-objective non-negative matrix decomposition models [33], graph regularization models [34], multi-objective constraint models [35], etc., where they decompose the objective matrix into the product of two low-rank matrices with other constraints, and solve the problem within a certain matrix rank bound. However, in practice, matrix decomposition methods typically requires pre-setting or estimating the matrix rank. This can increase the subjectivity of model parameter selection and, as a result, impact the accuracy of the complementary results. Later, the introduction of spatio-temporal correlation has inspired the accuracy enhancement of data recovery methods based on matrix complementation, such as the low rank and differential smoothing based recovery method (LRDS) proposed by Mao et al. [36] The differential smoothing prior of time-varying graph signals is introduced into the field of time-varying graph signal analysis, which enhances the accuracy of the data recovery. Yu et al. [19] use the spatio-temporal correlation as a regularization constraint term added to the hidden feature analysis model to make the data recovery both highly accurate and robust at the same time.

Table 1
Symbols and notations.

Symbols	Notations
X	Original data matrix, where each row corresponds to a day and each column corresponds to a time slot.
M	Ideal recovery matrix
S	Sparse error matrix
L	Sparse column error matrix
G	Gaussian noise matrix
$\hat{X}$	Original data matrix after column deletion in MOSDR-1
M_1, S_1, G_1	Ideal recovery matrix, error matrix and Gaussian noise matrix in MOSDR-1
$\bar{X}$	Raw data matrix after column backfilling and time pre-interpolation in MOSDR-2
$\bar{X}^{(i)}$	The i th anomaly column backfilled in $\bar{X}$
M_2, S_2, G_2	Ideal recovery matrix, error matrix and Gaussian noise matrix in MOSDR-2
D	Temporal differential matrix
K	Auxiliary matrix, $K = \bar{X}$
Y, Z_1, Z_2	Lagrange multiplicative matrix
W	weighting matrix
m, n	Number of rows and columns of X
t	Number of missing columns
D_τ	Singular value thresholding operator
S_τ	Soft thresholding operator
τ	Threshold value
μ	Weighting of equation constraints
λ_1, γ_1	Parameter weights of MOSDR-1
$\lambda_2, \gamma_2, \beta$	Parameter weights of MOSDR-2
$\ \cdot\ _*$	Nuclear norm
$\ \cdot\ _F$	Frobenius norm
$\ \cdot\ _1$	L1 norm
$\mathcal{R}_l$	Set of missing data indexes in the original observation matrix
$\mathcal{R}_c$	Set of data indexes for missing columns in the original observation matrix

Remark. However, in many cases, another dataset from a nearby sensor is not available (e.g., when budgetary constraints limit the deployment of duplicate sensors or when a nearby sensor is damaged) or the sensor's topology information is difficult to obtain. In such cases, the research question shifts from “how to use data from nearby sensors” to “how to use sensor data collected from the same sensor station to predict missing values”, which is inherently more challenging. Therefore, in this paper, we consider data recovery by assuming that data can only be obtained from one sensor station and by combining data records with their neighboring values using only temporal correlation.

3. Preliminaries

This section begins by clarifying the notation used throughout the paper. It then introduces the theory of matrix complementation before presenting the problem formulation of the paper.

3.1. Symbols and notations

Table 1 summarizes the notation and corresponding notes used in this paper.

3.2. Fundamentals of matrix completion

The matrix complementation problem can be mathematically modeled as the following constrained optimization model [31]:

$$\begin{aligned} \min_X \text{rank}(X) \\ \text{s.t. } P_\Omega(X) = P_\Omega(M) \end{aligned} \quad (1)$$

where $M \in \mathbb{R}^{m \times n}$ is the observed matrix of rank $r \ll \min(m, n)$ and some of the elements are known. X is the recovered low-rank matrix. $\Omega \subset \{1, 2, 3, \dots, m\} \times \{1, 2, 3, \dots, n\}$ denotes the set of subscript indices of

the observed elements of the matrix M . P_Ω is the orthogonal projection operator, which is defined as:

$$P_\Omega(M_{ij}) = \begin{cases} M_{ij}, & (i, j) \in \Omega \\ 0, & (i, j) \notin \Omega \end{cases} \quad (2)$$

Due to the presence of rank functions in Eq. (1), solving the problem becomes NP-hard [37]. Therefore, instead of using non-convex rank functions, the nuclear norm is utilized to relax the standard matrix-completion problem into the following convex-constrained optimization model:

$$\begin{aligned} \min_X \|X\|_* \\ \text{s.t. } P_\Omega(X) = P_\Omega(M) \end{aligned} \quad (3)$$

where $\|\cdot\|_*$ denotes the nuclear norm of the matrix.

3.3. Energy internet measurement data composition

In the Energy Internet environment, the integrity of measurement data is crucial for the stable operation and reliability assessment of the system. In this paper, we define a matrix $X_{m \times n}$ as the Energy Internet measurement data matrix, which consists of m time slots for n consecutive days, and each entry x_{ij} represents the measurement data for a specific date and a specific time slot. Considering the complexity of the Energy Internet measurement data, we can understand its constituent data matrix X as a superposition of the ideal matrix M , the error matrix S , the column error matrix L , and the Gaussian noise matrix G , as shown in Fig. 2.

The specific analyses are shown as follows:

(1) Energy Internet measurement data can be influenced by various factors, including environment, temperature, electricity price, and season. These data exhibit periodic distribution over time with a fixed regularity. Therefore, we can use regularly spaced measurement data to construct an ideal matrix under the low-rank assumption, where $\text{Rank}(M) \ll \min(m, n)$, in which $\text{Rank}(M)$ denotes the rank of M .

(2) During the process of acquiring measurement data, it is common to encounter recording errors and random data loss. Additionally, various network attacks, such as intrusion attacks, denial-of-service attacks, malware attacks, and physical attacks, may also result in transient power anomalies. However, as an adversary, the attacker has limited ability to attack additional measurement points. Therefore, any data anomalies caused by these attacks will be represented as discrete non-zero elements in the data matrix X , forming a sparse error matrix S . The error matrix S reflects missing, erroneous, and transient anomalies in the measurement data, which significantly impacts the accurate modeling and analysis of energy data.

(3) In real-world attack scenarios, attackers may also use Denial-of-Service (DoS) attacks to overload or crash a component of the power system or a network resource, resulting in a service interruption. This can cause power equipment to malfunction, resulting in a long-lasting anomaly. This prolonged anomaly is evident in the data as a significant deviation of one column of X from the other columns, resulting in a column error matrix L .

(4) During the process of data acquisition and transmission, noise is inevitably introduced, with Gaussian noise being the most common type. This noise is characterized by low amplitude and high density, and the presence of a Gaussian noise matrix G can have an impact on data recovery.

4. Proposed MOSDR model

4.1. MOSDR algorithm framework

To avoid the negative impact of long time missing data, combined with the analysis of the structure of the energy Internet measurement data above, we design an innovative shaping two-stage data recovery

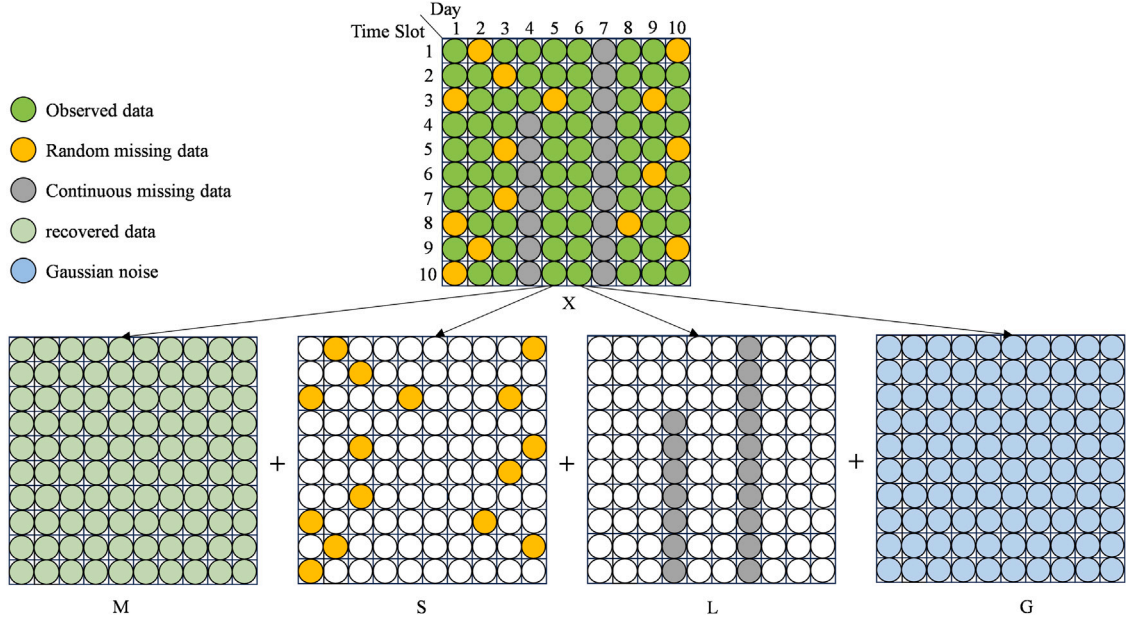


Fig. 2. Low-rank decomposition of measurement data matrix.

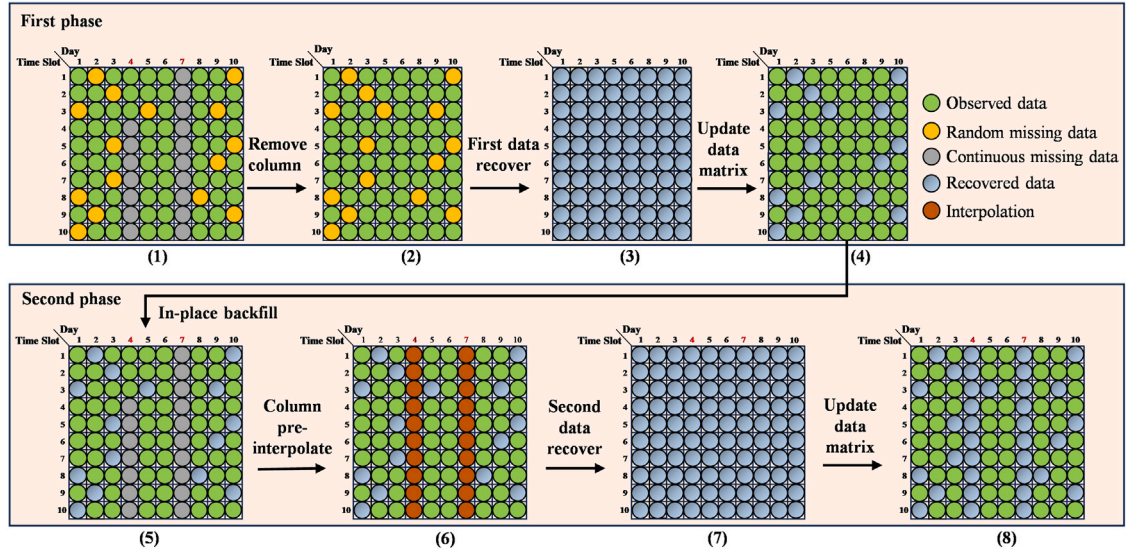


Fig. 3. MOSDR two stage data recovery scheme.

method MOSDR based on multi-norm joint optimization. The method utilizes the low-rank property of the data matrix and the temporal correlation to recover the data in phases, and the steps are shown in Fig. 3.

First phase: Remove t columns of long-term abnormal data, denoted as $x_a, x_b, \dots, x_t (t \ll n)$ from the observed data matrix X . This minimizes the negative impact on subsequent recovery accuracy. After this filtering step, we can construct an initially cleaned data matrix $\hat{X}_{m \times (n-t)}$. matrix $\hat{X}$ is a superposition of the ideal data matrix M_1 , the sparse anomaly data matrix S_1 , and the Gaussian noise matrix G_1 , that is $\hat{X} = M_1 + S_1 + G_1$. As previously stated, the low-rank component M_1 is represented by the nuclear norm in the objective function. The sparsity

of component S_1 is expressed using the L1 norm. Gaussian noise is characterized by high density and low amplitude, and is represented using the Frobenius norm. Based on these concepts, we propose the objective function for the first stage of data recovery:

$$\begin{cases} \min_{M_1, S_1, G_1} \|M_1\|_* + \lambda_1 \|S_1\|_1 + \gamma_1 \|G_1\|_F^2 \\ \text{s.t. } \hat{X} = M_1 + S_1 + G_1 \end{cases} \quad (4)$$

where λ_1, γ_1 is the parameter.

Second phase: In Eq. (4), we produce the first-stage recovery matrix $M_1 = [m_1^{(1)}, m_2^{(1)}, \dots, m_{n-t}^{(1)}]$, which serves as the matrix to be further recovered in the second stage. Initially, the abnormal columns removed

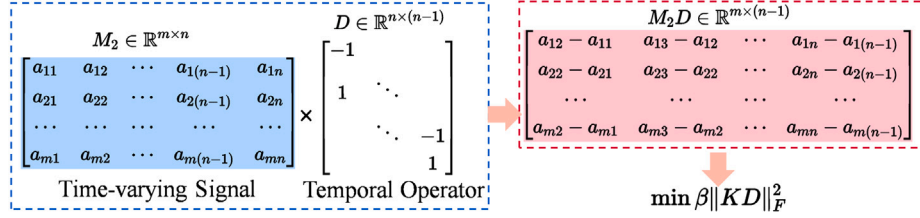


Fig. 4. Temporal smoothness schematic.

in the first stage are locally filled back within M_1 , resulting in matrix $\bar{X} = [\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n]$. Here, $m_i^{(1)}, i \in (1, 2, \dots, n-t)$ denotes the i th column vector of M_1 , and $\bar{x}_i, i \in (1, 2, \dots, n)$ denotes the i th column vector of $\bar{X}$. As measurement data typically exhibit strong temporal correlations within certain time intervals, this paper performs adjacent column mean imputation for the abnormal columns in B according to Eq. (5), where $\bar{X}^{(k)}$ represents the k th abnormal column filled back from $\bar{X}$, $k \in (a, b, \dots, t)$. Apart from the k anomalous columns in $\bar{X}^{(k)}$, the remainder of $\bar{X}$'s columns have been largely restored in the initial stage, specifically $[m_1^{(1)}, m_2^{(1)}, \dots, m_{n-t}^{(1)}]$. Hence, these columns are not subjected to further processing in the subsequent stage.

$$\bar{X}^{(k)} = \begin{cases} (\bar{X}^{(k-1)} + \bar{X}^{(k+1)})/2, \text{ other} \\ \bar{X}^{(2)}, k = 1 \\ \bar{X}^{(n-1)}, k = n \end{cases} \quad (5)$$

Meanwhile, inspired by the literature [36], the time differential operator D is introduced as follows.

$$D = \begin{bmatrix} -1 & & & \\ 1 & \ddots & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{bmatrix}_{n \times (n-1)} \quad (6)$$

As shown in Fig. 4, the time difference matrix of M_2 is M_2D , and $M_2D = [m_1^{(2)} - m_0^{(2)}, m_2^{(2)} - m_1^{(2)}, \dots, m_n^{(2)} - m_{n-1}^{(2)}]$. Normally, time series data does not exhibit significant fluctuations. Therefore, in the time difference matrix M_2D , only a few element values are expected to be very large, while most of the other values are relatively small. The L2 norm can improve the model's generalization ability and avoid overfitting. To achieve temporal smoothing of each data sequence and minimize data differences between adjacent time points, this paper proposes minimizing the Frobenius paradigm of M_2D and introducing an auxiliary matrix K , where $K = M_2$. Thus, the second-stage data recovery objective function is proposed as follows:

$$\begin{cases} \min_{M_2, S_2, G_2} \|M_2\|_* + \lambda_2 \|S_2\|_1 + \gamma_2 \|G_2\|_F^2 + \beta \|KD\|_F^2 \\ \text{s.t. } \bar{X} = M_2 + S_2 + G_2 \\ M_2 = K \end{cases} \quad (7)$$

where $\lambda_2, \gamma_2, \beta$ is the parameter.

4.2. ADMM-based optimization algorithm for solving

The low-rank matrix recovery model established by Eqs. (4) and (7) is an equationally constrained multi-objective optimization problem. The Alternating Direction Multiplier Method (ADMM) [38] method can decompose constrained optimization problems into smaller subproblems, thereby solving them iteratively and efficiently. Therefore, we uses the ADMM to seek the optimal solution to Eqs. (4) and (7).

Definition 1. Singular value thresholding operator D_τ : For a given matrix $A \in \mathbb{R}^{m \times n}$ of rank r and any threshold $\tau > 0$, let $A = U\Sigma V^T$ represent the singular value decomposition of matrix A , where $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{n \times r}$ are orthogonal matrices, $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$, and σ_i denotes the singular values of A . Then, the singular value thresholding operator $D_\tau(\bullet)$ can be defined as follows:

$$D_\tau(A) = UD_\tau(\Sigma)V^T = U \text{diag}(\max(0, \sigma_i - \tau))V^T \quad (8)$$

Definition 2. Soft thresholding operator S_τ : For a given matrix $A \in \mathbb{R}^{m \times n}$ and any threshold $\tau > 0$, the soft thresholding operator S_τ is defined as:

$$S_\tau(A) = \text{sgn}(A_{i,j}) \max\{0, |A_{i,j}| - \tau\} \quad (9)$$

where $A_{i,j}$ represents the element in the i th row and j th column of matrix A , and $\text{sgn}(A_{i,j})$ denotes the sign function of $A_{i,j}$.

4.2.1. First phase MOSDR-1

By constructing the augmented generalized Lagrangian function of the first stage model, shown in Eq. (4), the equationally constrained optimization problem is converted to an unconstrained optimization problem:

$$\begin{aligned} \min_{M_1, S_1, G_1} & \|M_1\|_* + \lambda_1 \|S_1\|_1 + \gamma_1 \|G_1\|_F^2 \\ & + \left\langle Y, \hat{X} - M_1 - S_1 - G_1 \right\rangle \\ & + \frac{\mu}{2} \|\hat{X} - M_1 - S_1 - G_1\|_F^2 \end{aligned} \quad (10)$$

where Y denotes the Lagrange multiplier matrix, μ is the constraint weight, and $(\bullet)$ is the inner product operation of the matrix. In Eq. (10), alternate iterations solve for M_1, S_1, G_1 .

(1) Update M_1 and fix S_1, G_1 . The objective function with respect to M_1 is then:

$$\begin{aligned} \arg \min_{M_1} & \|M_1\|_* + \left\langle Y, \hat{X} - M_1 - S_1 - G_1 \right\rangle + \frac{\mu}{2} \|\hat{X} - M_1 - S_1 - G_1\|_F^2 \\ = \arg \min_{M_1} & \frac{1}{\mu} \|M_1\|_* + \frac{1}{2} \|M_1 - \left(\hat{X} - S_1 - G_1 + \frac{1}{\mu} Y \right)\|_F^2 \end{aligned} \quad (11)$$

Eq. (11) has a closed-form solution [39], $M_{1(k+1)} = UD_\tau(\Sigma)V^T$. U, Σ, V is the factorization matrix for the singular value decomposition of the matrix $(\hat{X} - S_1 - G_1 + \frac{1}{\mu} Y)$, where $\tau = 1/\mu$.

(2) Update S_1 and fix M_1, G_1 . The objective function with respect to S_1 is then:

$$\begin{aligned} \arg \min_{S_1} & \lambda_1 \|S_1\|_1 + \left\langle Y, \hat{X} - M_1 - S_1 - G_1 \right\rangle + \frac{\mu}{2} \|\hat{X} - M_1 - S_1 - G_1\|_F^2 \\ = \arg \min_{S_1} & \frac{\lambda_1}{\mu} \|S_1\|_1 + \frac{1}{2} \|S_1 - \left(\hat{X} - M_1 - G_1 + \frac{1}{\mu} Y \right)\|_F^2 \end{aligned} \quad (12)$$

Eq. (12) has a closed-form solution [40], $S_{1(k+1)} = S_\tau(\hat{X} - M_1 - G_1 + \frac{1}{\mu} Y)$, where $\tau = \lambda_1/\mu$.

(3) Update G_1 and fix M_1, S_1 . The objective function with respect to G_1 is then:

$$\begin{aligned} & \arg \min_{G_1} \gamma_1 \|G_1\|_F^2 + \langle Y, \hat{X} - M_1 - S_1 - G_1 \rangle \\ & + \frac{\mu}{2} \|\hat{X} - M_1 - S_1 - G_1\|_F^2 \\ & = \arg \min_{G_1} \frac{\gamma_1}{\mu} \|G_1\|_F^2 + \frac{1}{2} \|G_1 - \left(\hat{X} - M_1 - S_1 + \frac{1}{\mu} Y\right)\|_F^2 \end{aligned} \quad (13)$$

Eq. (13) is an unconstrained optimization problem with double F-paradigm, and its analytical solution can be obtained directly by completing the square method. Let $B = \hat{X} - M_1 - S_1 + \frac{1}{\mu} Y$,

$$\begin{aligned} & \frac{\gamma_1}{\mu} \|G_1\|_F^2 + \frac{1}{2} \|G_1 - B\|_F^2 \\ & = \sum_{i=1,2,\dots,m} \sum_{j=1,2,\dots,n-t} \frac{\gamma_1}{\mu} g_{1(ij)}^2 + \frac{1}{2} \left(g_{1(ij)}^2 - 2g_{1(ij)}b_{ij} + b_{ij}^2 \right) \\ & = \sum_{i=1,2,\dots,m} \sum_{j=1,2,\dots,n-t} \left(\frac{\gamma_1}{\mu} + \frac{1}{2} \right) g_{1(ij)}^2 - g_{1(ij)}b_{ij} + \frac{1}{2} b_{ij}^2 \\ & = \sum_{i=1,2,\dots,m} \sum_{j=1,2,\dots,n-t} \left(\frac{2\gamma_1 + \mu}{2\mu} \right) \left(g_{1(ij)} - \frac{\mu}{2\gamma_1 + \mu} b_{ij} \right)^2 + \left[\frac{1}{2} - \frac{\mu}{2(2\gamma_1 + \mu)} \right] b_{ij}^2 \end{aligned} \quad (14)$$

where $g_{1(ij)}, b_{ij}$ are the elements of the i th row and j th column of the matrix G_1, B , respectively. The quadratic coefficient $\frac{2\gamma_1 + \mu}{2\mu}$ is constantly positive, so an optimum solution exists when $g_{1(ij)} = \frac{\mu}{2\gamma_1 + \mu} b_{ij}$. That is,

$$G_{1(k+1)} = \frac{\mu}{2\gamma_1 + \mu} B = \frac{\mu}{2\gamma_1 + \mu} \left(\hat{X} - M_1 - S_1 + \frac{1}{\mu} Y \right).$$

(4) Update $Y, Y_{k+1} = Y + \mu (\hat{X} - M_1 - S_1 - G_1)$.

The first phrase of MOSDR is shown in Algorithm 1.

Algorithm 1: MOSDR-1

Input: Original data matrix X , t missing columns, parameters λ_1, γ_1, μ , maximum iterations $k_{\max}$;

Output: Ideal recovery matrix M_1 , Sparse error matrix S_1 , Gaussian noise matrix G_1 ;

- 1 initialization: $\hat{X} \leftarrow$ Removing t missing columns
 $x_a, x_b, \dots, x_t (t \ll n)$ from X , $M_{1(0)} \leftarrow \hat{X}$, $S_{1(0)} \leftarrow 0$, $G_{1(0)} \leftarrow 0$,
 $Y_0 \leftarrow 0$ and determine the value of λ_1, γ_1, μ and $k_{\max}$;
 - 2 **while** $k < k_{\max}$ and not converge **do**
 - 3 $M_{1(k+1)} \leftarrow UD_{\tau}(\Sigma) V^T$;
 - 4 $S_{1(k+1)} \leftarrow S_{\tau} \left(\hat{X} - M_{1(k+1)} - G_{1(k)} + \frac{1}{\mu} Y_k \right)$;
 - 5 $G_{1(k+1)} \leftarrow \frac{\mu}{2\gamma_1 + \mu} \left(\hat{X} - M_{1(k+1)} - S_{1(k+1)} + \frac{1}{\mu} Y_k \right)$;
 - 6 $Y_{k+1} \leftarrow Y_k + \mu \left(\hat{X} - M_{1(k+1)} - S_{1(k+1)} - G_{1(k+1)} \right)$;
 - 7 **end**
-

4.2.2. Second phase MOSDR-2

By constructing the augmented generalized Lagrangian function of the second stage model, shown in Eq. (7), the equationally constrained optimization problem is converted to an unconstrained optimization problem:

$$\begin{aligned} & \min_{M_2, S_2, G_2, K} \|M_2\|_* + \lambda_2 \|S_2\|_1 + \gamma_2 \|G_2\|_F^2 + \beta \|KD\|_F^2 \\ & + \langle Z_1, \bar{X} - M_2 - S_2 - G_2 \rangle + \langle Z_2, M_2 - K \rangle \\ & + \frac{\mu_2}{2} \|\bar{X} - M_2 - S_2 - G_2\|_F^2 \\ & + \frac{\mu_2}{2} \|M_2 - K\|_F^2 \end{aligned} \quad (15)$$

where Z_1 and Z_2 denotes the Lagrange multiplier matrix, μ_2 is the constraint weight, and $\langle \cdot \rangle$ is the inner product operation of the matrix. In Eq. (15), alternate iterations solve for M_2, S_2, G_2, K .

$$\begin{cases} M_{2(k+1)} = \arg \min_{M_2} \|M_2\|_* + \langle Z_1, \bar{X} - M_2 - S_2 - G_2 \rangle + \langle Z_2, M_2 - K \rangle \\ \quad + \frac{\mu_2}{2} \|\bar{X} - M_2 - S_2 - G_2\|_F^2 + \frac{\mu_2}{2} \|M_2 - K\|_F^2 \\ S_{2(k+1)} = \arg \min_{S_2} \lambda_2 \|S_2\|_1 + \langle Z_1, \bar{X} - M_2 - S_2 - G_2 \rangle \\ \quad + \frac{\mu_2}{2} \|\bar{X} - M_2 - S_2 - G_2\|_F^2 \\ G_{2(k+1)} = \arg \min_{G_2} \gamma_2 \|G_2\|_F^2 + \langle Z_1, \bar{X} - M_2 - S_2 - G_2 \rangle \\ \quad + \frac{\mu_2}{2} \|\bar{X} - M_2 - S_2 - G_2\|_F^2 \\ K_{k+1} = \arg \min_K \beta \|KD\|_F^2 + \langle Z_2, M_2 - K \rangle + \frac{\mu_2}{2} \|M_2 - K\|_F^2 \end{cases} \quad (16)$$

where $M_{2(k+1)}, S_{2(k+1)}, G_{2(k+1)}, K_{k+1}$ are the $(k+1)$ th iteration values of M_2, S_2, G_2, K respectively. In the same way as the previous section, the equation can be further rewritten as:

$$\begin{cases} M_{2(k+1)} = \arg \min_{M_2} \frac{1}{\mu_2} \|M_2\|_* + \|M_2 - \nabla_M\|_F^2 \\ S_{2(k+1)} = \arg \min_{S_2} \frac{\lambda_2}{\mu_2} \|S_2\|_1 + \frac{1}{2} \|S_2 - \nabla_S\|_F^2 \\ G_{2(k+1)} = \arg \min_{G_2} \gamma_2 \|G_2\|_F^2 + \langle Z_1, \bar{X} - M_2 - S_2 - G_2 \rangle \\ \quad + \frac{\mu_2}{2} \|\bar{X} - M_2 - S_2 - G_2\|_F^2 \\ K_{k+1} = \arg \min_K \beta \|KD\|_F^2 + \langle Z_2, M_2 - K \rangle \\ \quad + \frac{\mu_2}{2} \|M_2 - K\|_F^2 \end{cases} \quad (17)$$

(1) $M_{2(k+1)} = UD_{\tau}(\Sigma) V^T$. U, Σ, V is the factorization matrix for the singular value decomposition of the matrix ∇_M , where $\nabla_M = \left(\frac{\bar{X} - S_2 - G_2 + (Z_1/\mu_2) + K - (Z_2/\mu_2)}{2} \right)$, $\tau = \frac{1}{2\mu_2}$.

(2) $S_{2(k+1)} = S_{\tau}(\nabla_S)$, where $\nabla_S = \frac{\bar{X} - M_2 - G_2 + (Z_1/\mu_2)}{2}$.

(3) $G_{2(k+1)} = \frac{\mu_2}{2\gamma_2 + \mu_2} \left(\bar{X} - M_2 - S_2 + \frac{Z_1}{\mu_2} \right)$.

(4) The objective function $\arg \min_K \beta \|KD\|_F^2 + \langle Z_2, M_2 - K \rangle + \frac{\mu_2}{2} \|M_2 - K\|_F^2$ with respect to K is a continuously derivable convex function, for which the derivation yields $2\beta K D D^T - Z_2 + \mu_2 (M_2 - K) = 0$. Then, the optimal solution is

$$K_{k+1} = (Z_2 + \mu_2 M_2) (2\beta D D^T + \mu_2 I)^{-1} \quad (18)$$

(5) Finally, the update rule for Lagrange multipliers is:

$$\begin{aligned} Z_1 &= Z_1 + \mu_2 (\bar{X} - M_2 - S_2 - G_2) \\ Z_2 &= Z_2 + \mu_2 (M_2 - K) \end{aligned} \quad (19)$$

The second phrase of MOSDR is shown in Algorithm 2.

5. Experiments

All experiments in this paper were conducted on a personal computer equipped with an Intel(R) Core(TM) i7-8550U CPU @ 1.80 GHz with 16.0 GB RAM and a 64-bit Windows 10 operating system. The algorithms in the experiments are developed based on Python 3 environment. The relevant code for this paper is available from <https://github.com/tawry/MOSDR.git>. The following experiments are designed to answer the following research questions (RQs):

- RQ1: How does MOSDR perform compared to other methods as the proportion of random missing data and the number of missing columns increases?
- RQ2: How does MOSDR perform in different datasets?

Algorithm 2: MOSDR-2

Input: The original data matrix $\bar{X}$ of the second stage obtained after pre-interpolation according to Eq. (5), parameters $\lambda_2, \gamma_2, \mu_2, \beta$, maximum iterations $k_{\max}$;
Output: Ideal recovery matrix M_2 , Sparse error matrix S_2 , Gaussian noise matrix G_2 , auxiliary matrix K ;

- 1 initialization: $M_{2(0)} \leftarrow \bar{X}, S_{2(0)} \leftarrow 0, G_{2(0)} \leftarrow 0, Z_{1(0)} \leftarrow 0, Z_{2(0)} \leftarrow 0$ and determine the value of $\lambda_2, \gamma_2, \mu_2, \beta$ and $k_{\max}$;
- 2 **while** $k < k_{\max}$ **and not converge** **do**
- 3 $M_{2(k+1)} \leftarrow U D_{\tau}(\Sigma) V^T$;
- 4 $S_{2(k+1)} \leftarrow S_{\tau} \left(\frac{\bar{X} - M_{2(k+1)} - G_{2(k)} + (Z_{1(k)}/\mu_2)}{2} \right)$;
- 5 $G_{2(k+1)} \leftarrow \frac{\mu_2}{2\gamma_2 + \mu_2} \left(\bar{X} - M_{2(k+1)} - S_{2(k+1)} + \frac{Z_{1(k)}}{\mu_2} \right)$;
- 6 $K_{k+1} \leftarrow (Z_{2(k)} + \mu_2 M_{2(k+1)}) (2\beta D D^T + \mu_2 I)^{-1}$;
- 7 $Z_{1(k+1)} \leftarrow Z_{1(k)} + \mu_2 \left(\bar{X} - M_{2(k+1)} - S_{2(k+1)} - G_{2(k+1)} \right)$;
- 8 $Z_{2(k+1)} \leftarrow Z_{2(k)} + \mu_2 (M_{2(k+1)} - K_{k+1})$;
- 9 **end**

Table 2

Properties of the experimental datasets.

Abbreviations	Datasets	Instances	Features
PQT	Real-life power quality transients	400	50
NTR	Nantong-Resident	96	159
PCTC	Power consumption of Tetouan city	144	364
AEP	Appliances Energy Prediction	144	137

5.1. Datasets

Four benchmark datasets are selected to conduct the experiments. Table 2 summarizes their properties.

(1) Real-life Power Quality Transients (PQT) [41]: This dataset contains power quality transient events recorded in the real grid caused by various grid disturbances such as voltage dips and spikes, etc. The power quality data is collected at a sampling frequency of 20 Hz for 50 cycles, with 400 data points sampled in each cycle.

(2) Nantong-Resident (NTR): This dataset is sourced from Nantong Power Supply Company, State Grid Corporation of China, and can be found at <https://github.com/dxdora/data-resource>. The dataset consists of electricity consumption data from residential households, with daily load data sampled at 15-min intervals and a feature size of 96. It provides a detailed record of the daily electricity consumption patterns of the residents, which is valuable for analyzing and predicting the behavioral patterns of household electricity consumption.

(3) Power Consumption of Tetouan City (PCTC): The dataset available at <https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city> records urban distribution grid consumption in Tetouan, Morocco, sampled at 10-min intervals. These data are valuable for studying electricity distribution and consumption patterns at an urban scale.

(4) Appliances Energy Prediction (AEP): this dataset collects energy consumption data from household appliances sampled at ten-minute intervals, which can provide data support for analyzing the energy efficiency of different appliances and daily energy use patterns, and is available at <https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction>.

To evaluate the performance of the MOSDR algorithm when dealing with incomplete datasets, we performed manual missing operations on these datasets. Specifically, we simulated real-world data loss or corruption by randomly removing segments of the complete dataset at varying missing rates ($R = 0.2, 0.4, 0.6, 0.8$) and also by selectively deleting portions of specific columns ($t = 2, 4, 6, 8, 10$). The missing rate is the proportion of missing data to the total number of data. As stated

in [42], unbiased estimation can be achieved by assigning zero weight to missing data when the data follow the random missing assumption. Thus, the weight of each element is expressed as:

$$W_{ij} = \begin{cases} 1, & \text{if } X_{ij} \text{ is observed} \\ 0, & \text{if } X_{ij} \text{ is missing} \end{cases} \quad (20)$$

5.2. Baselines

In order to show the performance of the proposed method in a fair and objective way, this paper compares the MOSDR method with several matrix complementation methods such as SVT [39], RPCA [18], LADMAP [43], LFA [44], PNMF [45], MIWAE [46] and GAIN [47]. The details of the methods compared are given below:

(1) SVT: The Singular Value Thresholding (SVT) algorithm is a method for approximating matrices that satisfy a set of convex constraints by minimizing the nuclear norm. The algorithm's core lies in applying a soft thresholding operation to sparse matrices to remove smaller singular values and enhance matrix sparsity. SVT has an important property where the rank of the obtained matrix is non-decreasing during the iteration process. This indicates that the algorithm tends to recover a solution closer to the rank of the original matrix with each iteration step.

(2) RPCA: Robust Principal Component Analysis (RPCA) is an efficient data analysis technique mainly used to extract low-rank structures from data containing noise or outliers. The basic principle is to decompose the data matrix into two parts: a low-rank matrix and a sparse matrix. The low-rank matrix captures the main structure of the data, while the sparse matrix contains outliers or noise. This decomposition is achieved by minimizing the nuclear norm of the low-rank matrix and the L1 norm of the sparse matrix. RPCA is particularly suitable for scenarios in which outliers or noise have a significant impact on the effectiveness of traditional Principal Component Analysis (PCA), and is effective in recovering clean data structures from dirty data.

(3) LADMAP: Linearized Alternating Direction Method with Adaptive Penalty (LADMAP) is a method used to solve convex optimization problems with linear constraints. It simplifies the solution of sub-problems in the traditional Alternating Direction Method (ADM) by linearizing the quadratic penalty term and introducing the nearest neighbor term. Unlike Robust Principal Component Analysis (RPCA), LADMAP is a more efficient approach. Additionally, the algorithm enhances convergence speed and guarantees global convergence by adaptively adjusting penalty parameters.

(4) LFA: Latent Factor Analysis (LFA) is a matrix decomposition-based method designed to recover lost data from sparse matrices. The LFA model assumes that the goal is to train two hidden factor matrices with a low-rank approximation obtained by their product. These factor matrices are trained by minimizing the sum of errors between the original and recovered matrices and by adding regularization terms.

(5) PNMF: Probabilistic Matrix Factorization (PMF) mines potential features by decomposing high-dimensional data matrices in a probabilistic framework. Unlike traditional matrix factorization methods, PMF introduces a probabilistic model to deal with the uncertainty and incompleteness of the data, usually assuming that the observed data are generated by the product of two low-rank matrices plus Gaussian noise. By maximizing the probabilistic likelihood function of the data, PMF can effectively recover useful information from sparse and noisy data.

(6) MIWAE: MIWAE is a deep latent variable model (DLVM) method for handling missing-at-random data. It is based on the Importance Weighted Autoencoder (IWAE) and trains the model by optimizing a tight lower bound of the log-likelihood of the observed data. Compared to the original IWAE, MIWAE does not introduce any additional computational overhead when dealing with missing data.

(7) GAIN: Generative Adversarial Imputation Nets (GAIN) is a generative adversarial network (GAN) model designed for handling missing data. Its core idea is to generate reasonable imputed values for missing data through adversarial learning.

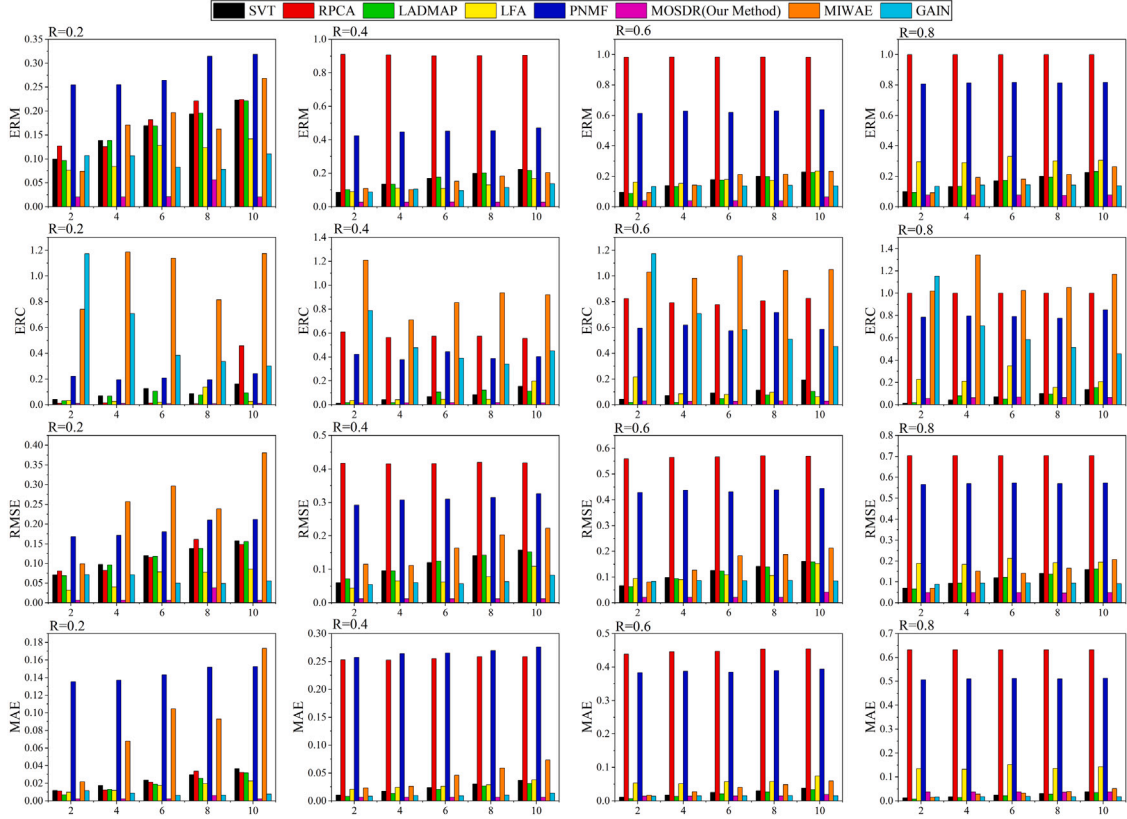


Fig. 5. PQT dataset: Comparison of ERM, ERC, RMSE, MAE metrics of SVT, RPCA, LADMAP, LFM, PNMF, MOSDR, MIWAE, GAIN methods with different random missing ratios ($R = 0.2, 0.4, 0.6, 0.8$) and different number of missing columns ($C = 2, 4, 6, 8, 10$).

5.3. Evaluation metrics

To compare the recovery performance of different methods and scenarios, we define two comparison metrics: Error Ratio on Missing data (ERM) and Error Ratio on Missing Column (ERC).

Definition 3. Let $\mathcal{R}_l$ denote the set of missing data indexes in the original observation matrix, and X_{ij} and $\hat{X}_{ij}$ ($i, j \in \mathcal{R}_l$) denote the original observation matrix and the recovery matrix, respectively.

Then, we call $\frac{\sqrt{\sum_{i,j \in \mathcal{R}_l} (X_{ij} - \hat{X}_{ij})^2}}{\sqrt{\sum_{i,j \in \mathcal{R}_l} X_{ij}^2}}$ the error ratio on missing data (ERM).

In the paper, we use *ERM* to measure the error after recovery of random missing data in the original observation matrix.

Definition 4. Let $\mathcal{R}_c$ denotes the set of data indexes for missing columns in the original observation matrix, and X_{ij} and $\hat{X}_{ij}$ ($i, j \in \mathcal{R}_c$) denote the original observation matrix and the recovery matrix, respectively.

Then, we call $\frac{\sqrt{\sum_{i,j \in \mathcal{R}_c} (X_{ij} - \hat{X}_{ij})^2}}{\sqrt{\sum_{i,j \in \mathcal{R}_c} X_{ij}^2}}$ the error ratio on missing column (ERC).

In the paper, we use *ERC* to measure the error after recovery of randomly missing columns in the original observation matrix.

In addition, we used the root mean square error (RMSE) and mean absolute error (MAE) as evaluation criteria. The specific formulas are as follows:

$$RMSE = \sqrt{\frac{1}{|\mathcal{R}|} \sum_{i,j \in \mathcal{R}} (X_{ij} - \hat{X}_{ij})^2} \quad (21)$$

$$MAE = \frac{1}{|\mathcal{R}|} \sum_{i,j \in \mathcal{R}} |X_{ij} - \hat{X}_{ij}| \quad (22)$$

where $|\mathcal{R}|$ refers to the total number of data observed in the observation matrix.

5.4. Results and discussion

5.4.1. Answering RQ1

This paper comprehensively evaluates four different datasets as presented in the above paper. The experimental results are shown in Figs. 5 to 8, corresponding to the results of PQT, NTR, PCTC, and AEP datasets, respectively. To ensure comparability between the datasets and eliminate the influence of different magnitudes and ranges, we normalized the four datasets. This guarantees that the algorithm's performance is not biased by the magnitude of the original data. We examine the performance of the MOSDR algorithm in comparison with several matrix completion methods, SVT, RPCA, LADMAP, LFM, PNMF, MIWAE, GAIN for different random missing ratios ($R = 0.2, 0.4, 0.6, 0.8$) as well as different number of missing columns ($C = 2, 4, 6, 8, 10$). The quantitative analysis was carried out through two metrics, ERM and ERC, supplemented by RMSE and MAE as additional assessment metrics. Next we will analyze each of the four indicators.

• Analysis of *ERM*:

(1) As the proportion of missing data increases while holding the number of missing columns constant, MOSDR consistently demonstrates markedly lower ERM values compared to other methods in most cases. As depicted in Figs. 5 and 6, even when the missing proportion reaches as high as 0.8, MOSDR consistently maintains a relatively low ERM, whereas the ERM of other methods notably increases. In particular, the ERM of RPCA reaches 1, indicating complete failure of this method under high missing proportions.

• Analysis of *ERC*:

(1) From Figs. 5 to 8, it can be observed that, when the random missing proportion remains constant, MOSDR generally exhibits lower ERC compared to other methods as the number of missing columns increases, demonstrating its stability. MOSDR employs a

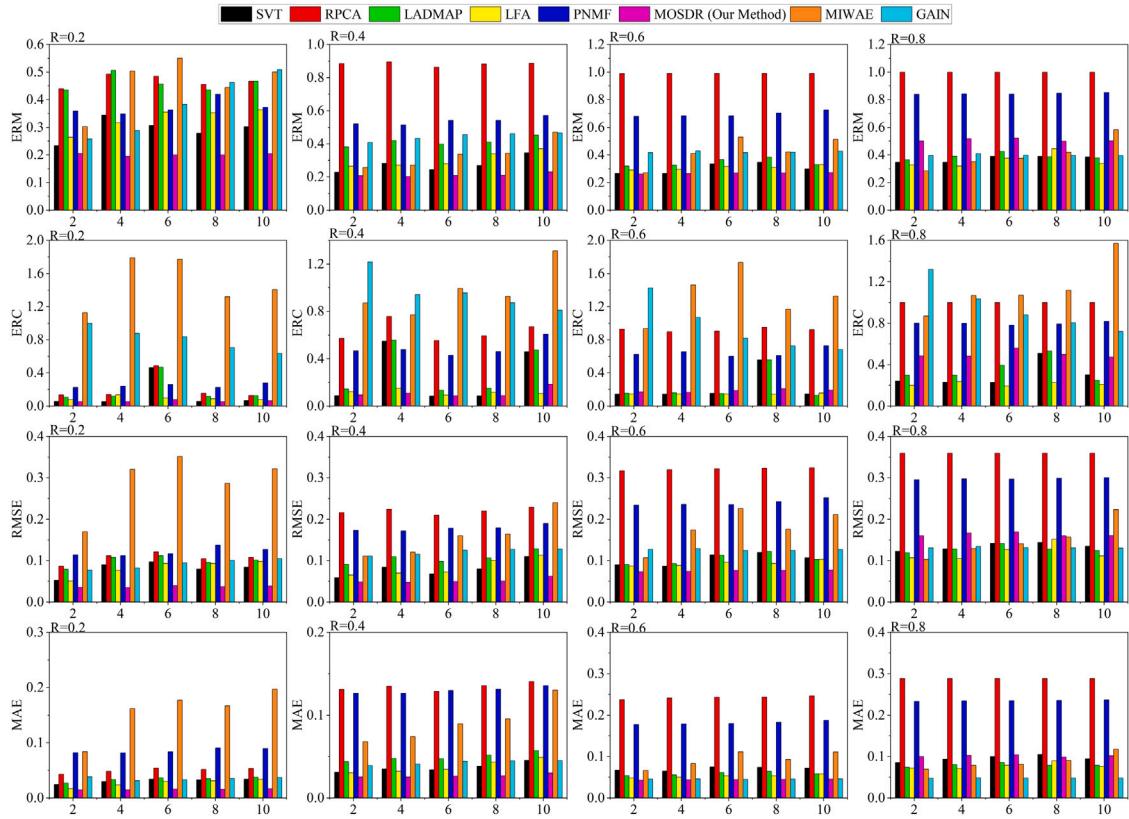


Fig. 6. NTR dataset: Comparison of ERM, ERC, RMSE, MAE metrics of SVT, RPCA, LADMAP, LFM, PNMF, MOSDR, MIWAE, GAIN methods with different random missing ratios ($R = 0.2, 0.4, 0.6, 0.8$) and different number of missing columns ($C = 2, 4, 6, 8, 10$).

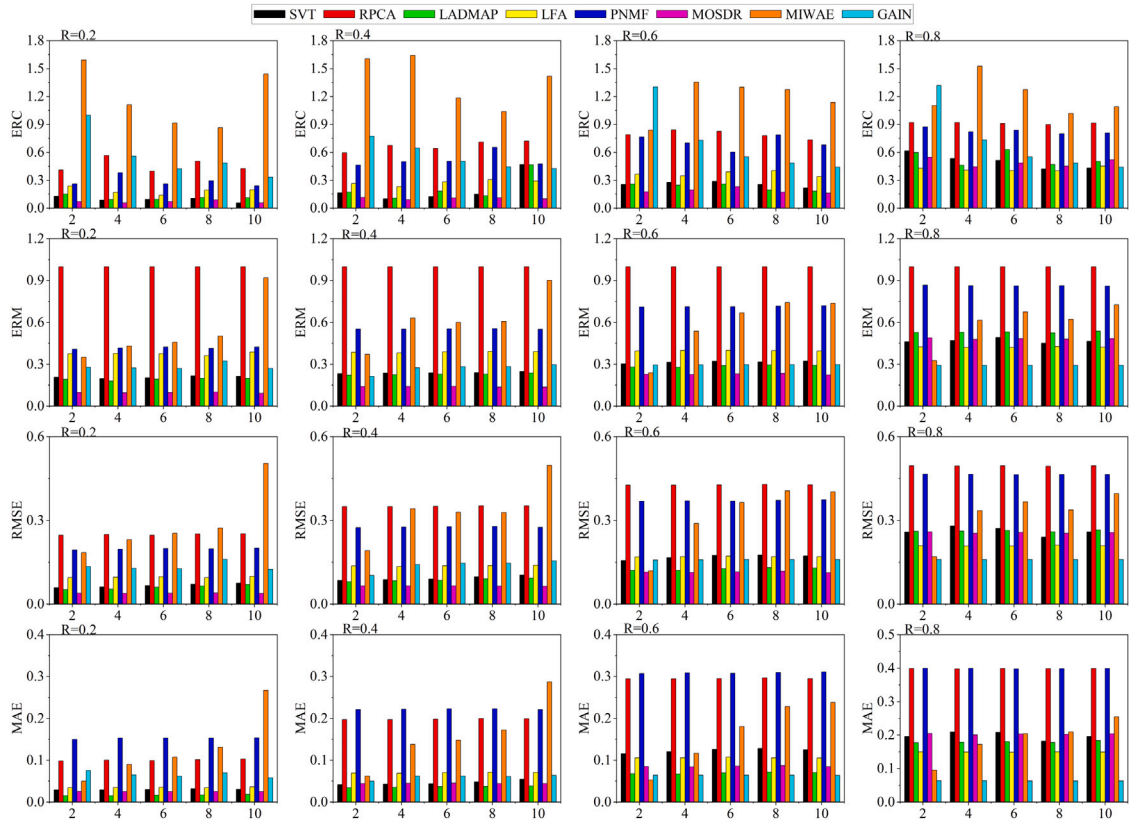


Fig. 7. PCTC dataset: Comparison of ERM, ERC, RMSE, MAE metrics of SVT, RPCA, LADMAP, LFM, PNMF, MOSDR, MIWAE, GAIN methods with different random missing ratios ($R = 0.2, 0.4, 0.6, 0.8$) and different number of missing columns ($C = 2, 4, 6, 8, 10$).

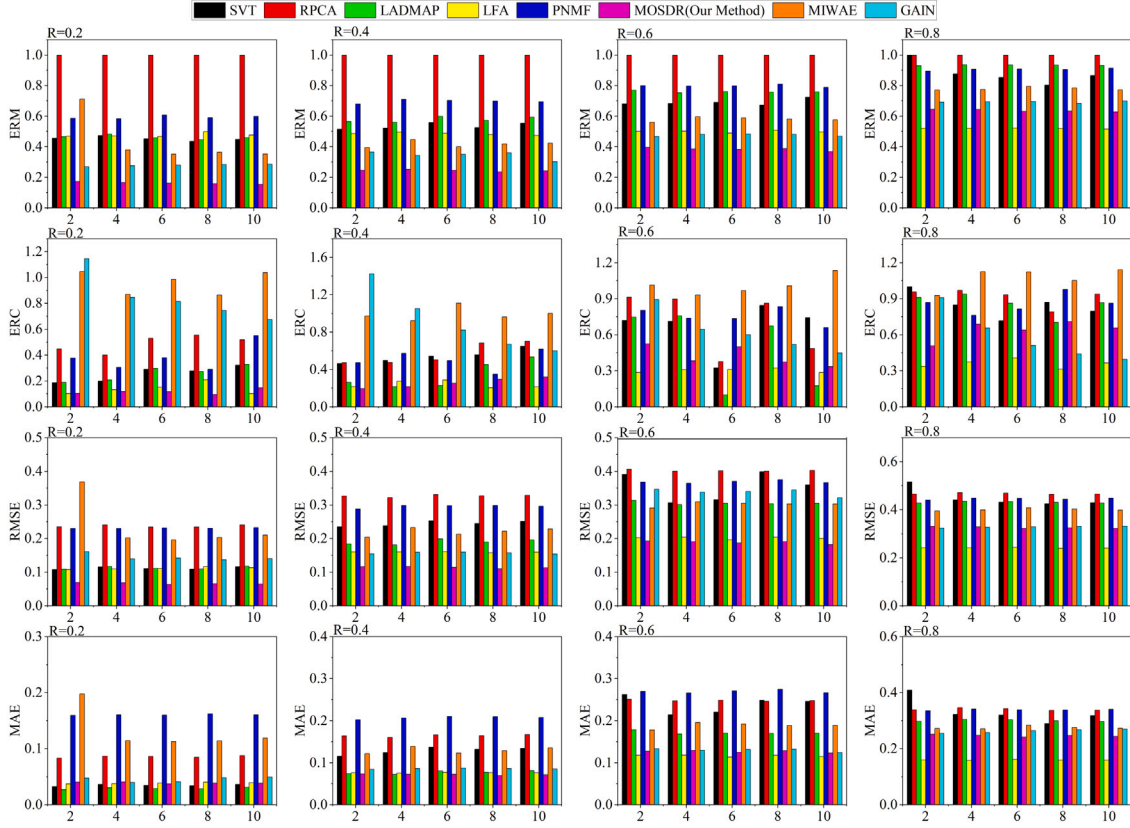


Fig. 8. AEP dataset: Comparison of ERM, ERC, RMSE, MAE metrics of SVT, RPCA, LADMAP, LFM, PNMf, MOSDR, MIWAE, GAIN methods with different random missing ratios ($R = 0.2, 0.4, 0.6, 0.8$) and different number of missing columns ($C = 2, 4, 6, 8, 10$).

phased processing mechanism that first removes the anomalous columns and then recovers the temporal correlation of the data through a differencing operation. This approach not only captures the global distribution of the data, but also preserves the local details, which effectively reduces the error when facing structured missing data. As a result, MOSDR consistently maintains a low error value on the ERC metric.

(2) The common feature of SVT, RPCA, LADMAP, LFM, and PNMf is to recover data through matrix decomposition and optimization techniques, which are usually more effective in dealing with low-rank matrices and linear missing patterns, but underperform in facing complex missing patterns.

(3) MIWAE exhibits higher errors as the number of missing columns increases, which is mainly due to the model's simpler assumptions about the distribution of the latent variables, which reduces the ability to learn the latent variables at high missing rates, leading to insufficient interpolation accuracy.

(4) As the number of missing columns increases, GAIN shows a decreasing trend in the ERC metrics. This is mainly due to the fact that with only a few missing columns, GAIN may overfit the local information, leading to anomalies in the interpolated values. As the number of missing columns increases, the model completes each missing value with less reliance on specific local features and more on the overall distribution of the data. This reduces local overfitting and therefore error. Despite the significant reduction in ERC of GAIN, its overall performance is still not as good as MOSDR when dealing with complex missing column structures.

• Analysis of RMSE and MAE:

(1) The MOSDR algorithm exhibits low errors in terms of RMSE and MAE metrics, particularly at random missing ratios $R = 0.2$ and $R = 0.4$, where it outperforms other methods significantly. This result highlights the accuracy advantage of the MOSDR

algorithm, particularly when dealing with structured missingness, such as column missingness. At high missing ratios (e.g., $R = 0.8$), MOSDR still maintains good quality of recovery despite the performance degradation of all methods.

(2) The MOSDR algorithm consistently outperforms all other methods on all evaluation metrics. This is due to its ability to fully utilize not only the compositional characteristics of the energy Internet measurement data, but also the temporal correlation and structured characteristics of the data. Compared to traditional SVT and RPCA methods, MOSDR has a clear advantage in handling complex data structures and a large proportion of missing data. While the LADMAP method approaches the performance of MOSDR in some cases, it still lags behind in scenarios where columns are missing. LFM and PNMf, as methods based on matrix decomposition, show competitiveness on some datasets, but their performance declines when faced with more difficult missing patterns.

5.4.2. Friedman test (Answering RQ2)

The Friedman test is a non-parametric statistical test primarily used for comparing the performance of multiple algorithms across multiple datasets to determine if there are significant performance differences. Let K algorithms be compared on N datasets, and r_i denote the mean ordinal value of the i th algorithm. Then the mean and variance of r_i are $(K+1)/2$ and $(K^2+1)/12N$, respectively, and the variable $\Gamma_F = \frac{(N-1)\Gamma_{\chi^2}}{N(K-1)-\Gamma_{\chi^2}}$ obeys the F distribution with degrees of freedom $k-1$ and

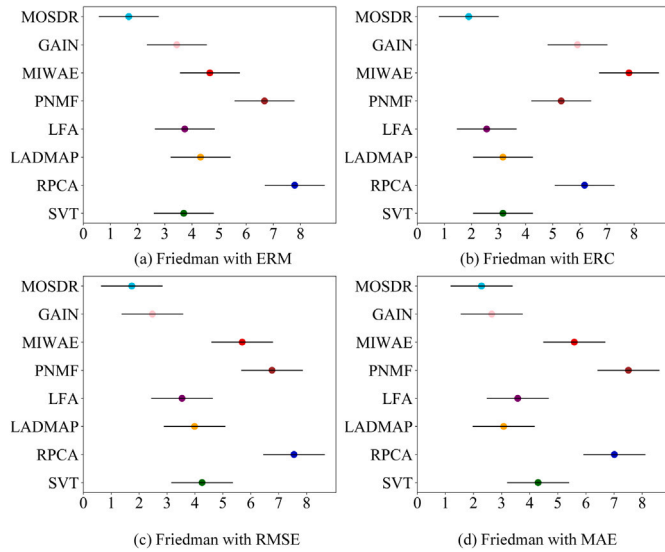
$$(k-1)(N-1), \text{ where } \Gamma_{\chi^2} = \frac{12N}{K(K+1)} \left(\sum_{i=1}^K r_i^2 - \frac{K(K+1)^2}{4} \right).$$

To verify the reliability of the experimental results that demonstrate the higher accuracy of the proposed MOSDR model compared to other algorithms across different datasets, we evaluated four performance metrics (ERM, ERC, RMSE, MAE) on four datasets (PQT, NTR, PCTC,

Table 3

Friedman test results.

Metrics	PQT	NTR	PCTC	AEP
	p-value	p-value	p-value	p-value
ERM	5.20E-20	3.24E-16	3.67E-20	2.07E-19
ERC	1.49E-20	2.71E-22	2.71E-22	9.44E-18
RMSE	9.85E-21	1.40E-18	1.13E-19	5.69E-21
MAE	3.50E-19	1.25E-18	2.31E-18	4.07E-20

**Fig. 9.** Friedman side view of the six models based on the Nemenyi test.

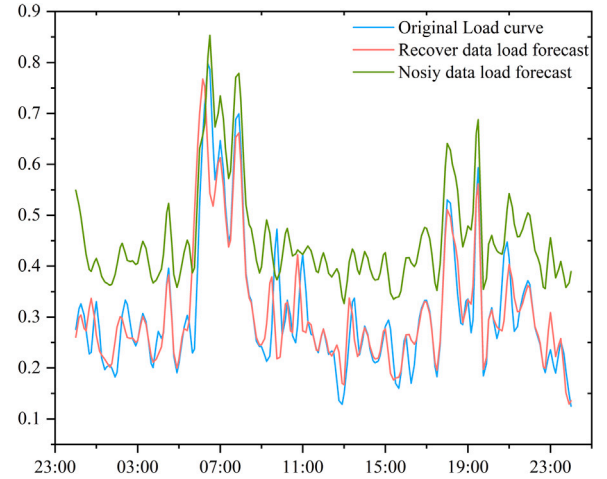
AEP). We conducted Friedman and Nemenyi post-hoc tests on the experimental results of the eight aforementioned models. The results of the Friedman test are presented in Table 3. The findings indicate that the p-values of the Friedman test for all datasets and metrics are significantly less than 0.05, indicating substantial performance differences among the various algorithms.

We further differentiated the performance of various algorithms using the Nemenyi post-hoc test. Fig. 9 illustrates the performance of the eight algorithms across all datasets for four different metrics. In the figure, the horizontal axis represents the ranks, the vertical axis represents the algorithm names, the dots indicate the average rank values, and the horizontal lines denote the confidence intervals of the ranks.

From Fig. 9, we can draw the following conclusions:

- The MOSDR algorithm excels across all four metrics (ERM, ERC, RMSE, MAE) across all datasets, achieving the highest average rank and exhibiting smaller confidence intervals, which demonstrates the algorithm's stable and superior performance in terms of recovery accuracy.
- LFA and GAIN shows some competitiveness in certain metrics, particularly RMSE and MAE, but overall, it still falls short compared to MOSDR. SVT performs the worst across all metrics, indicating that this algorithm does not have an advantage with these datasets and metrics.
- The performance of the other algorithms (PNMF, LADMAP, RPCA, MIWAE) falls between that of MOSDR and SVT, with specific performance varying depending on the metric and dataset.

Through the Nemenyi test and an integrated analysis of all datasets, we can clearly see that the proposed method consistently outperforms the seven compared methods across the four metrics: ERM, ERC, RMSE,

**Fig. 10.** The performance of load forecasting on data with anomalies and after recovery.**Table 4**

Load forecasting accuracy comparison.

Data	RMSE	MAE
Noisy data	0.1731	0.1581
Recover data	0.0908	0.0646

and MAE. This consistent performance advantage across datasets underscores the applicability and robustness of our method to various types and sizes of datasets.

5.4.3. Case study

In real-world energy internet scenarios, load forecasting plays a critical role. Yet, untreated anomalous raw data can significantly disrupt load forecasting models, resulting in erroneous forecasts. To assess the effect of anomaly data recovery on subsequent load forecasting, we conducted an experiment. We chose the NTR dataset, introducing 10% anomalies at random. We split the data into two sets: one with untreated anomalies and another with anomalies rectified by the MOSDR method. Subsequently, we applied the same linear regression model for load forecasting to both data sets.

Fig. 10 compares load forecasting performance on data with and without anomaly recovery. It shows that models trained on data containing anomalies have substantial inaccuracies, with predictions deviating greatly from actual loads and errors, such as RMSE and MAE listed in Table 4, increasing markedly. Conversely, forecasts based on MOSDR-recovered data align more closely with actual values, suggesting the data's enhanced reliability and accuracy for forecasting. This is due to the error propagation caused by anomalies, which can skew model parameters and, in time series models, affect subsequent predictions. Recovered data counters these effects, enabling the model to track load changes more accurately and thus enhance forecast precision and stability.

This case analysis underscores the importance of anomaly recovery for accurate subsequent load forecasting, highlighting the practical utility of the MOSDR method in energy internet data processing.

5.4.4. Discussion

The MOSDR method uses a two-stage optimization process with various matrix operations and constraints, which can be computationally expensive with large-scale data, especially for real-time processing

where efficiency is crucial. Enhancing the algorithm's computational efficiency for large and high-dimensional datasets is a key optimization area. Moreover, MOSDR presupposes precise anomaly detection and localization, yet in practice, these tasks are often intertwined. Inaccurate anomaly detection can compromise recovery outcomes. Integrating anomaly detection seamlessly with data recovery to form a cohesive framework is a significant practical challenge. We will continue to explore these areas to improve the practicality and adaptability of the method.

6. Conclusions

This paper proposes a two-stage data recovery method based on multinorm optimized structured data recovery (MOSDR) to address the issue of recovering abnormal measurement data in the context of the energy internet. The method relies on accurate detection and labeling of abnormal data and is based on the intrinsic characteristics of the measurement data. In the first stage, low-rank and sparse constraints are used to recover randomly missing data. In the second stage, MOSDR incorporates temporal correlation constraints into the objective function. This results in the smoothing of time-pre-interpolated column-missing data, which further enhances recovery accuracy. The model is efficiently solved using the ADMM. Experimental results on four real-world datasets demonstrate that MOSDR effectively recovers missing data and outperforms existing methods in key performance metrics such as ERM, ERC, RMSE, and MAE. MOSDR breaks through the limitations of traditional matrix completion techniques and accurately restores the data matrix, even when certain columns are completely empty. In future work, our aim is to improve the computational efficiency and scalability of the MOSDR algorithm. This will enable more efficient handling of larger datasets, expanding its application scope in the field of the energy internet and related domains.

CRedit authorship contribution statement

Song Deng: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Li Tao:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Dong Yue:** Writing – review & editing, Project administration, Funding acquisition. **Yi He:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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